

**INSTITUTE OF ENGINEERING AND TECHNOLOGY
DEVI AHILYA UNIVERSITY, INDORE**

B.E. 1 year (Sem-1), Mid Sem Test -1, OCTOBER 2023

Subject: Basic Electronics

Class: IT (A & B)

Code: IETRC4

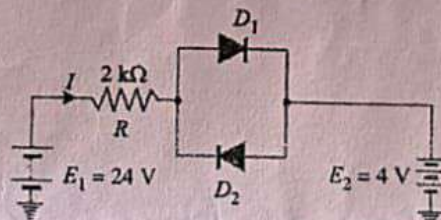
M. Marks: 20

Note: Attempt any four questions.

Q1. a) When two different materials with distinct fermi levels are brought into contact, what happens to the fermi levels and what is the resulting charge transfer mechanism at the interface? [2]

b) Consider a silicon PN junction diode with donor and acceptor concentration of $5 \times 10^{16} \text{ cm}^{-3}$ and $1 \times 10^{17} \text{ cm}^{-3}$ respectively. Calculate the built-in potential (V_{bi}) and depletion width (W) under zero bias. Discuss how changes in donor and acceptor concentrations would affect these values. Given $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$ and $\epsilon_s = 11.7$ [3]

Q2. a) Determine the current I in the circuit shown in figure. Assume the diodes to be silicon and forward resistance of the diodes be zero. [2]



b) Draw V-I characteristics of a PN junction diode. Explain why the current under reverse bias is almost independent of the applied voltage up to the critical voltage. [3]

Q3. a) Define the following terms:

1. Fermi level

2. Recombination

[2]

b) How do variations in semiconductor materials (e.g., silicon and germanium) impact the V-I characteristic of diodes? Discuss the differences in terms of voltage drop and performance. [3]

Q4. Determine the probability of finding an electron in the valence band, conduction band, and mid-gap energy level within an intrinsic semiconductor, and elaborate on how temperature influences the fermi level in such materials. [5]

Q5. Explain working mechanism of npn transistor. Draw Input and Output characteristics for CE configuration and mention active, saturation and cut-off regions. [5]